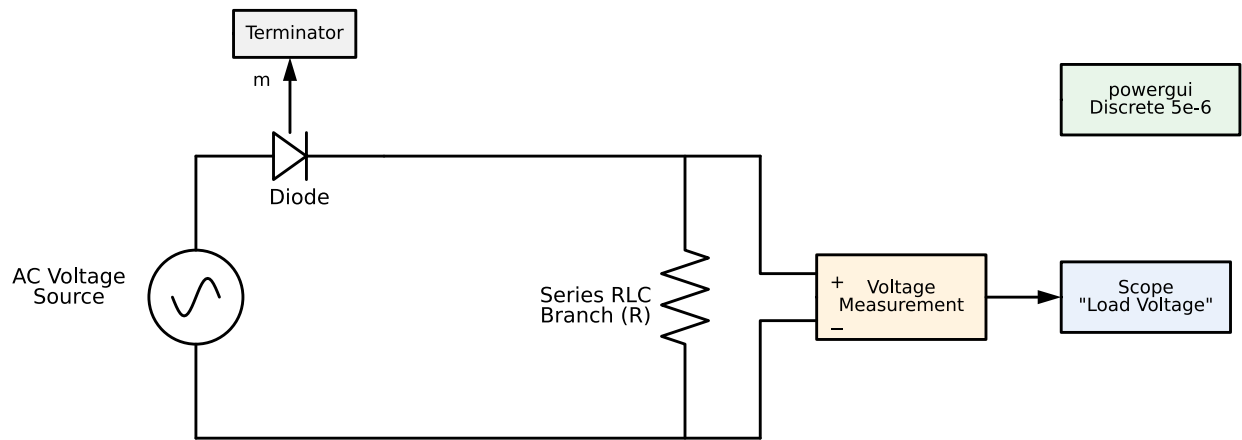


PE Lab — Final Circuit of Every Experiment

Left: circuit with every part labelled by its Simulink block name (blue boxes = signal blocks, orange = measurement blocks). Right: Block → SET. Signal names v, i = outputs of Voltage / Current Measurement (wire them directly or use Goto/From). MATLAB R2024b · Simscape Electrical → Specialized Power Systems.

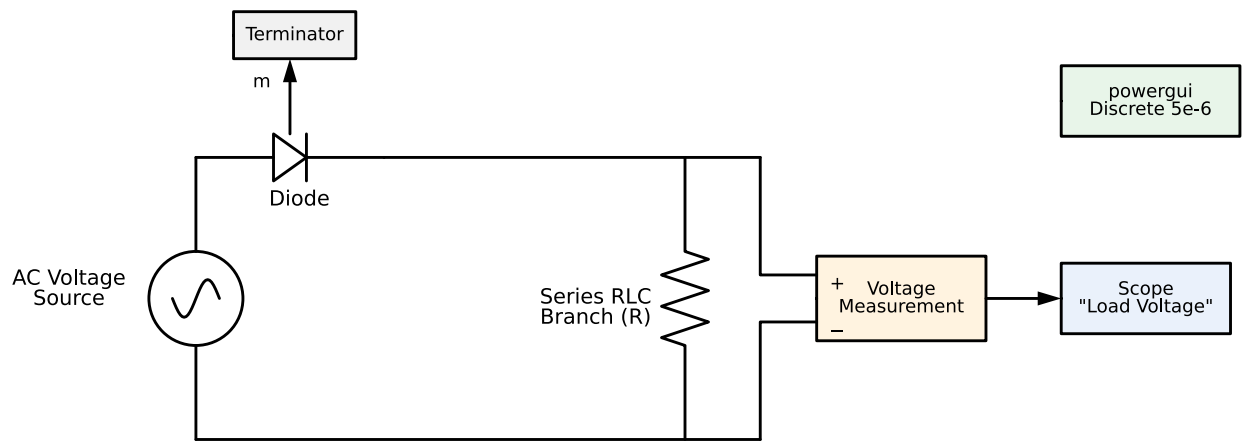
1.1 Single-phase half-wave diode rectifier (R load)



Block (ADD)	SET
AC Voltage Source	Peak amplitude 311 , Phase 0, Frequency 60
Diode	Snubber Rs 1e7
Series RLC Branch	Branch type R, Resistance 10
Voltage Measurement	—
Scope (rename "Load Voltage")	—
powergui	Discrete, 5e-6
Terminator	on unused m-ports

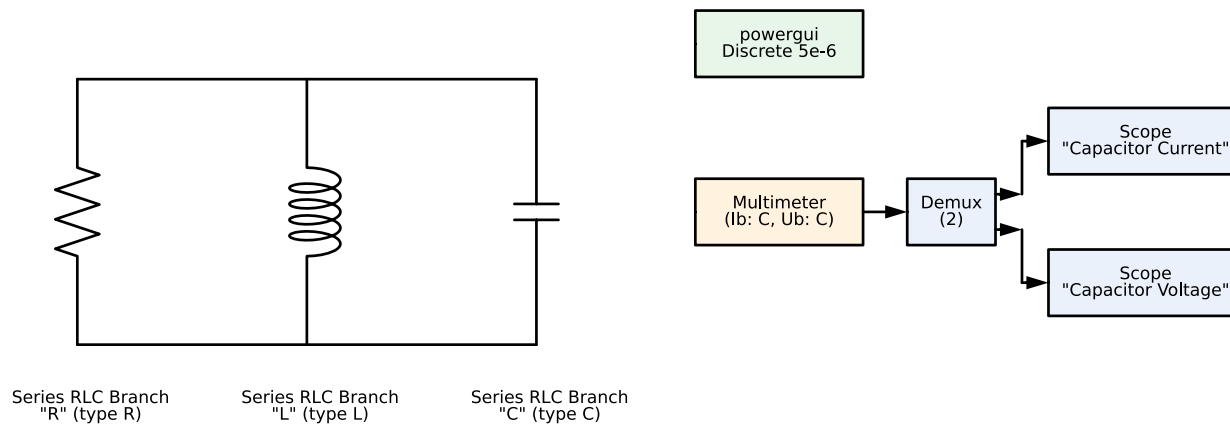
Stop time 100e-3

1.2 Measurement with the Oscilloscope (Scope) block



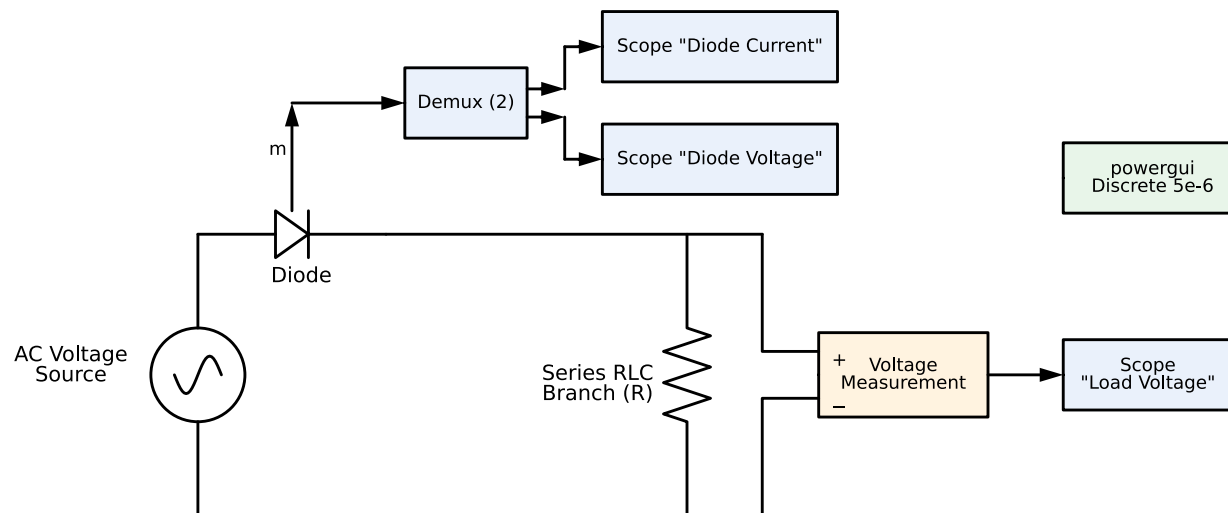
Block (ADD)	SET
(same circuit as 1.1)	—
Scope → Cursor Measurements	two cursors one period apart → $1/\Delta T = 60$ Hz
Scope → Zoom X	select one full steady-state cycle
Scope → Signal Statistics	read Max, Min, Mean , RMS
Expected	Max 310.2, Mean 98.71 V, RMS 155.1 V

1.3 Measurement with the Multimeter block (parallel RLC)



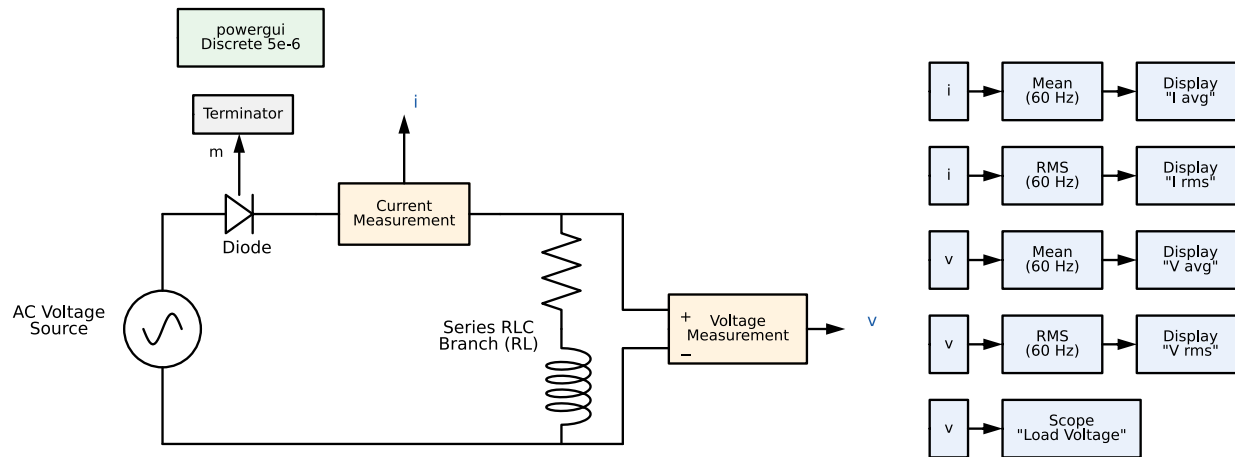
Block (ADD)	SET
Series RLC Branch "R"	Branch type R, 10
Series RLC Branch "L"	Branch type L, 250e-3, ☒ initial current 0.1
Series RLC Branch "C"	Branch type C, 2e-3, ☒ initial voltage 2, Measurements Branch voltage and current
Multimeter	select Ib: C, Ub: C → >>
Demux	Number of outputs 2
Scope ×2	"Capacitor Current", "Capacitor Voltage"
powergui	Discrete, 5e-6
Stop time 0.5	

1.4 Measurement port (diode current and voltage)



Block (ADD)	SET
AC Voltage Source	Peak amplitude 311, Phase 0, Frequency 60
Diode	Snubber Rs 1e7
Series RLC Branch	R, 10
Demux	2 (out1 = i_{AK} , out2 = v_{AK})
Scope ×3	"Diode Current", "Diode Voltage", "Load Voltage"
Voltage Measurement	—
powergui	Discrete, 5e-6

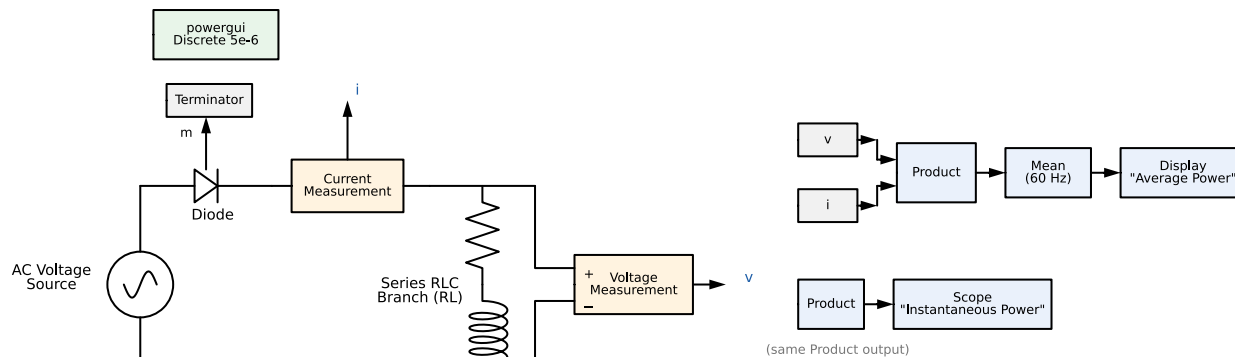
1.5 Mean and RMS blocks (half-wave, RL load)



Block (ADD)	SET
AC Voltage Source	Peak amplitude 100 , Phase 0, Frequency 60
Diode	Snubber Rs 1e7
Series RLC Branch	Branch type RL, R 100 , L 0.1
Current Measurement	—
Voltage Measurement	—
Mean ×2 (Sensors and Measurements)	Fundamental frequency 60
RMS ×2	Fundamental frequency 60
Display ×4	—
Scope	“Load Voltage”
powergui	Discrete, 5e-6
Terminator	on unused m-ports

Stop time 834e-3 · Expected I_{avg} 0.3041 A, I_{rms} 0.4685 A, V_{avg} 30.4 V, V_{rms} 49.74 V

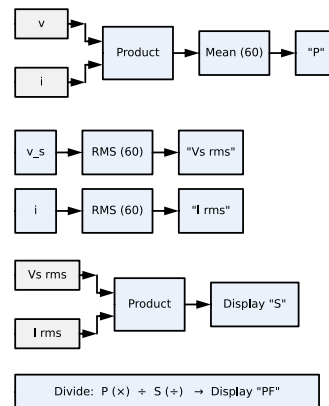
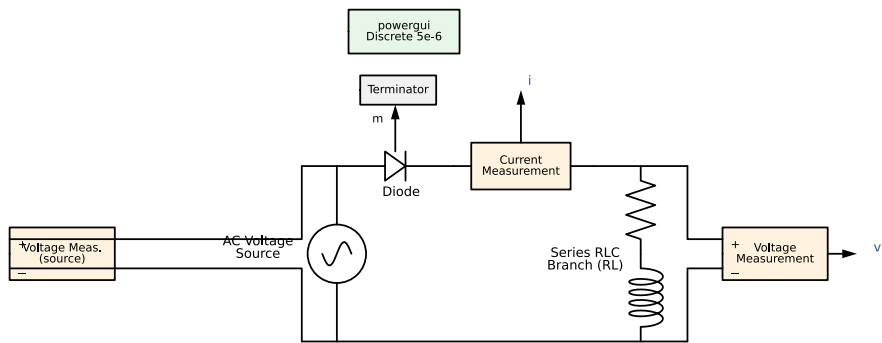
1.6 Instantaneous power and average power



Block (ADD)	SET
(circuit of 1.5)	same settings
Product	Number of inputs 2 (v, i)
Mean	Fundamental frequency 60
Display	“Average Power”
Scope	“Instantaneous Power”
powergui	Discrete, 5e-6

Expected $P = 21.95$ W

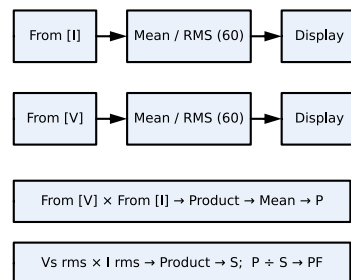
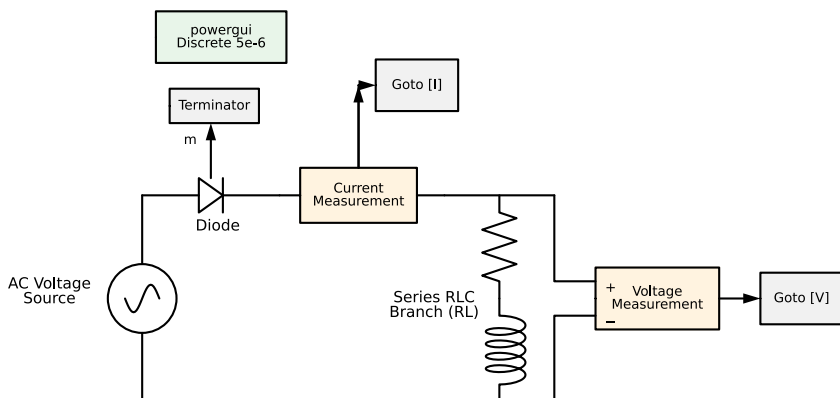
1.7 Apparent power and power factor



Block (ADD)	SET
(circuit of 1.5)	same settings
Voltage Measurement (2nd)	across the AC source
RMS ×2	60 (source V, load I)
Product ×2	$P = \text{mean}(v \cdot i)$; $S = V_{s,rms} \cdot I_{rms}$
Mean	60
Divide	top $\times = P$, bottom $\div = S$
Display ×3	"P", "S", "PF"
powergui	Discrete, 5e-6

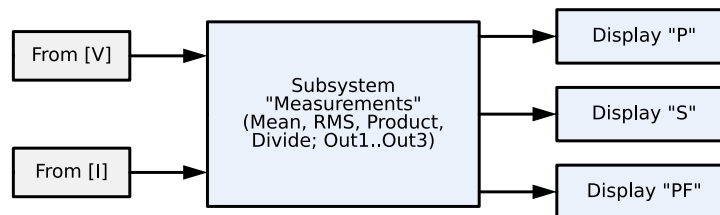
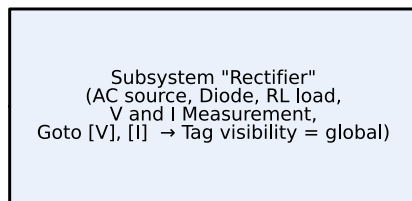
Expected $S = 33.13$ VA, $PF = 0.6626$

2.1 Goto and From blocks



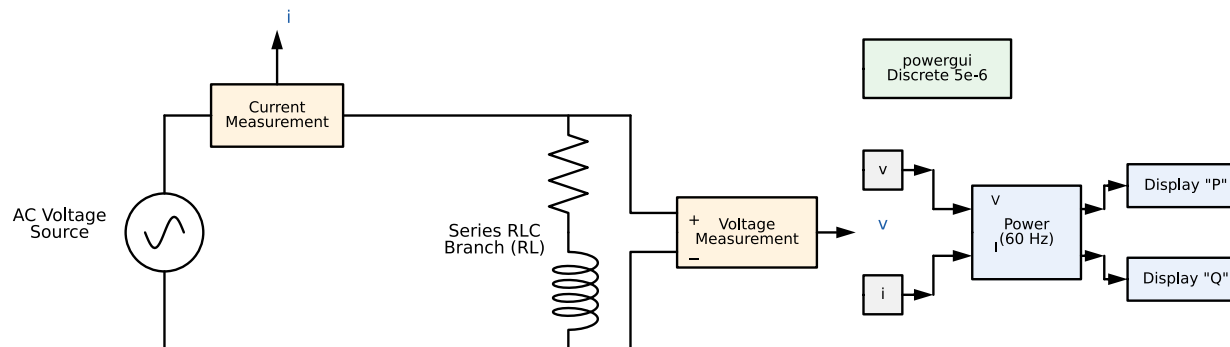
Block (ADD)	SET
(circuit of 1.7)	same settings
Goto ×2	Goto tag V and I
From (as many as needed)	Goto tag V or I
Mean / RMS / Product / Divide / Display	as 1.7
powergui	Discrete, 5e-6

2.2 Making a subsystem



Block (ADD)	SET
Select circuit + Ctrl+G	Subsystem "Rectifier"
Goto (inside subsystem)	Tag visibility global
Select measurement blocks + Ctrl+G	Subsystem "Measurements", rename Out1... Out3
From ×2	V , I
Display ×3	P, S, PF

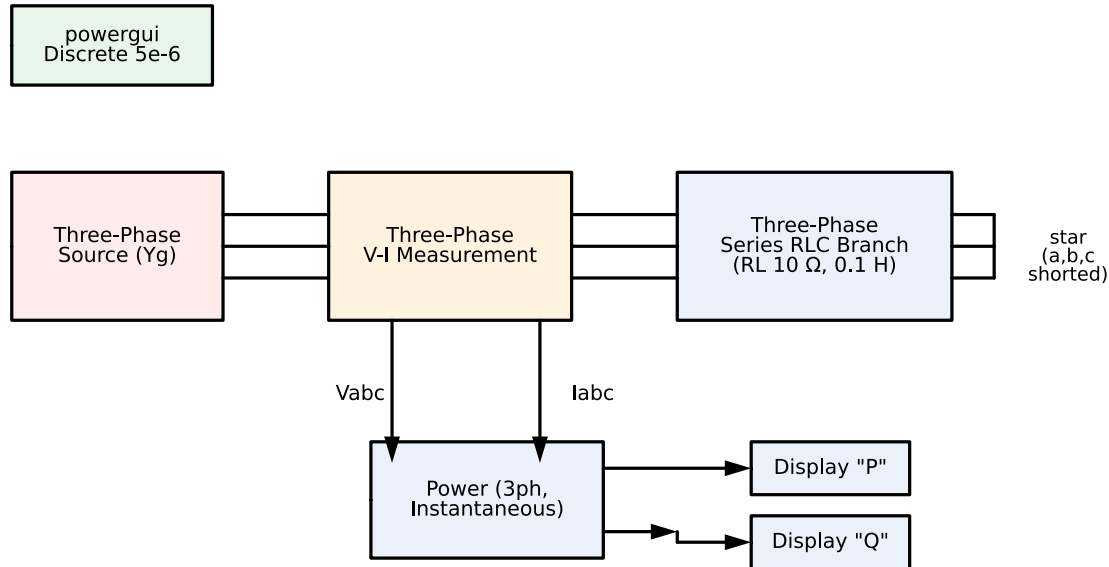
2.3a Power block — single-phase linear RL load



Block (ADD)	SET
AC Voltage Source	Peak amplitude 100 , Phase 0, Frequency 60
Series RLC Branch	RL, 100 , 0.1
Current Measurement	—
Voltage Measurement	—
Power (Sensors and Measurements)	Fundamental frequency 60
Display ×2	"P", "Q"
powergui	Discrete, 5e-6

Expected P 43.78 W, Q 16.5 VAR

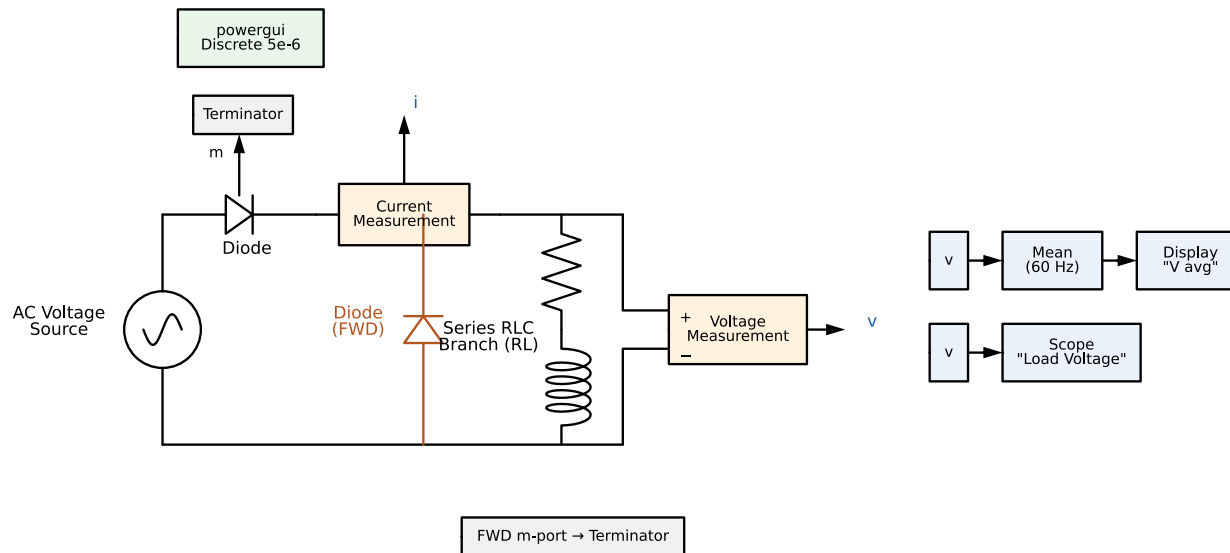
2.3b Power block — three-phase linear load



Block (ADD)	SET
Three-Phase Source	Configuration Yg, Phase-to-phase Vrms 207.8461, Freq 60, ☐ Internal
Three-Phase V-I Measurement	Voltage phase-to-ground, Current yes
Three-Phase Series RLC Branch	RL, R 10, L 0.1 (short a, b, c together)
Power (3ph, Instantaneous)	—
Display ×2	"P", "Q"
powergui	Discrete, 5e-6

Expected P 284 W, Q 1071 VAR, PF 0.2563

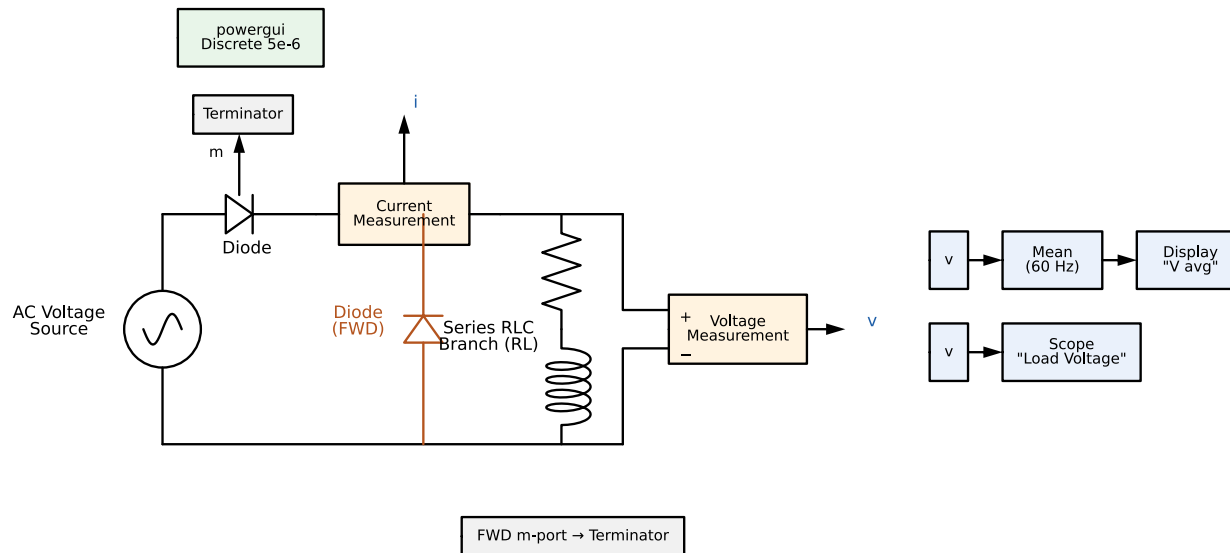
2.4 Freewheeling diode



Block (ADD)	SET
AC Voltage Source	Peak amplitude 100, Phase 0, Frequency 60
Diode ×2 (main + FWD)	Snubber Rs 1e7; FWD anode bottom, cathode top
Series RLC Branch	RL, 100, 0.1
Current Measurement	—
Voltage Measurement	—
Mean	60
Display	"V avg"
Scope	"Load Voltage"
powergui	Discrete, 5e-6
Terminator	on unused m-ports

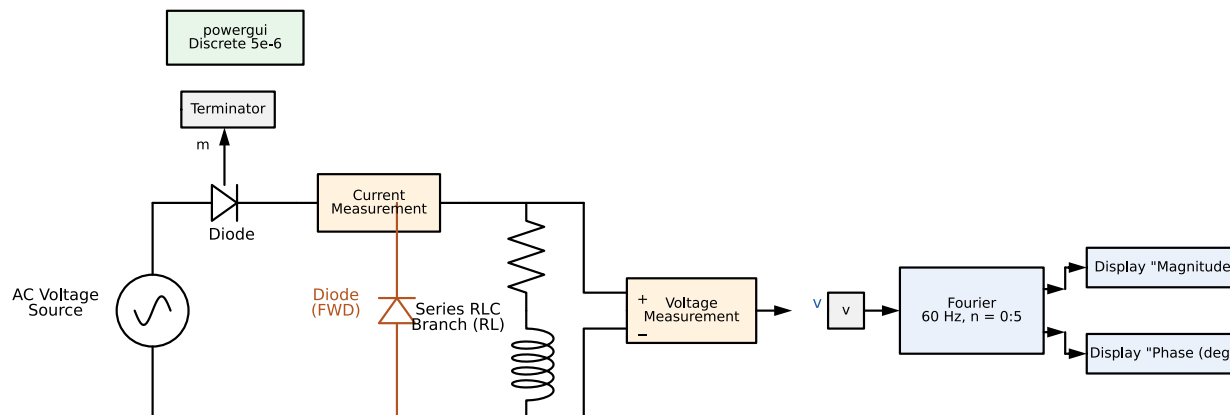
Stop time 834e-3 · Expected V_{avg} 31.25 V

2.5 Disabling a block



Block (ADD)	SET
(circuit of 2.4)	same settings
FWD → right-click	Comment Out (Ctrl+Shift+X) → grey
Run	V_{avg} back to 30.4 V
FWD → right-click	Uncomment to enable

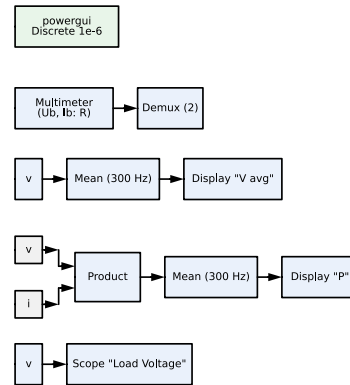
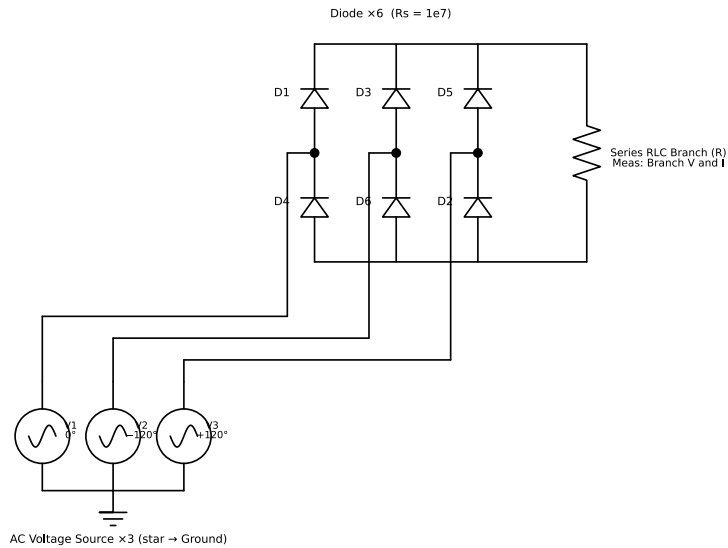
2.6 Fourier block (harmonics of load voltage)



Block (ADD)	SET
(circuit of 2.4)	same settings
Fourier	Fundamental 60 , Harmonic n 0:5 , Initial [0 , 0] , Sample time 5e-6
Display ×2	"Magnitude", "Phase (deg)" (enlarge)
powergui	Discrete, 5e-6

Expected magnitudes: DC 31.83, n1 50, n2 21.22, n4 4.24 V

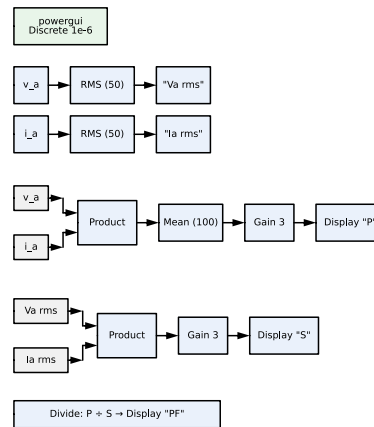
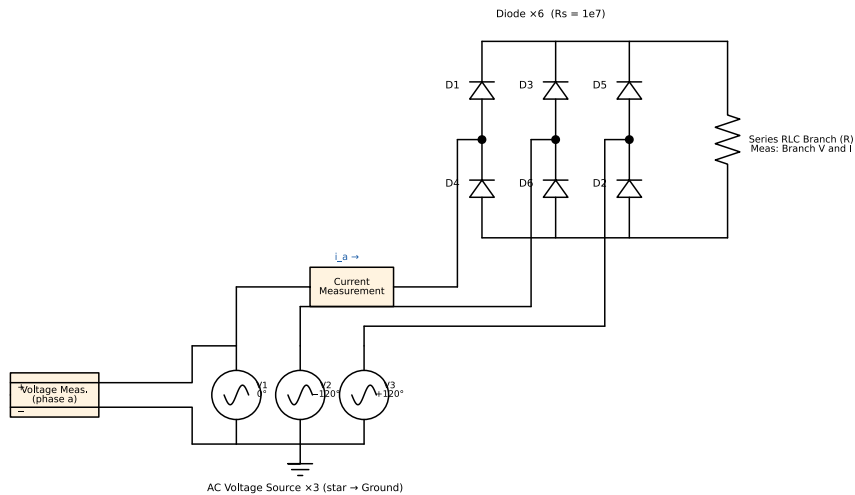
2.7 Three-phase diode rectifier



Block (ADD)	SET
AC Voltage Source ×3	Peak $220 \cdot \sqrt{2}$, Freq 50 , Phase 0 / -120 / 120
Ground	bottom of all 3 sources
Diode ×6	Snubber Rs 1e7
Series RLC Branch	R 10 , Measurements Branch voltage and current
Multimeter	Ub, Ib of load
Demux	2
Mean ×2	300
Product	2
Display ×2, Scope	—
powergui	Discrete, 1e-6

Stop time 0.1 · Expected V_{avg} 512.7 V, P 26.33 kW

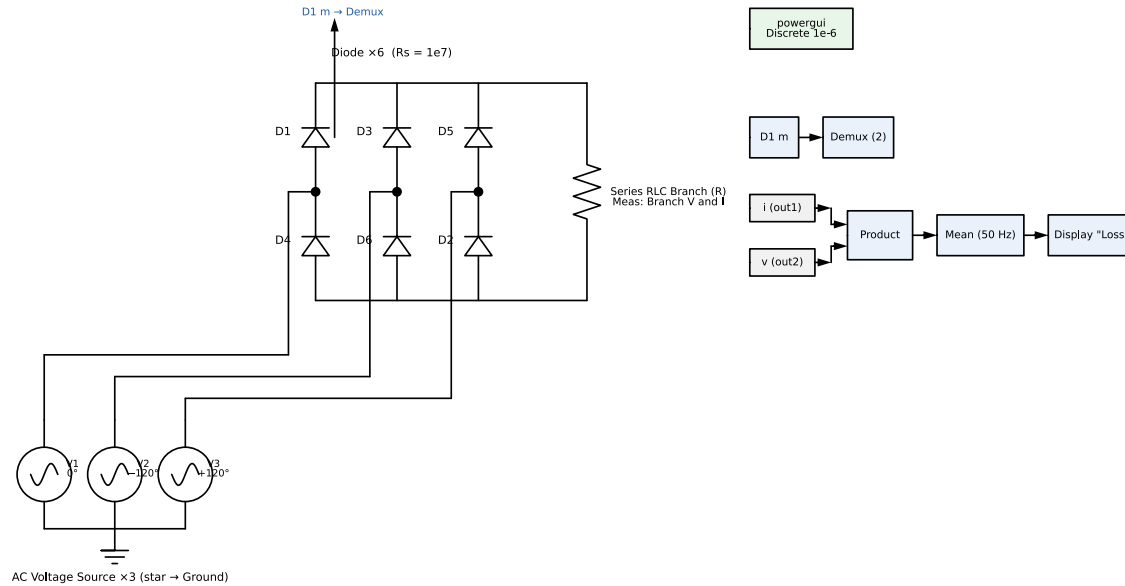
2.8 Power factor of the three-phase rectifier



Block (ADD)	SET
(circuit of 2.7)	same settings
Current Measurement	in series with V1 (phase a)
Voltage Measurement	across V1
RMS ×2	50
Product ×2, Mean	Mean 100
Gain ×2	3
Divide	$P \div S$
Display ×3	P, S, PF
powergui	Discrete, 1e-6

Expected P 26.42 kW, S 27.64 kVA, PF 0.9558

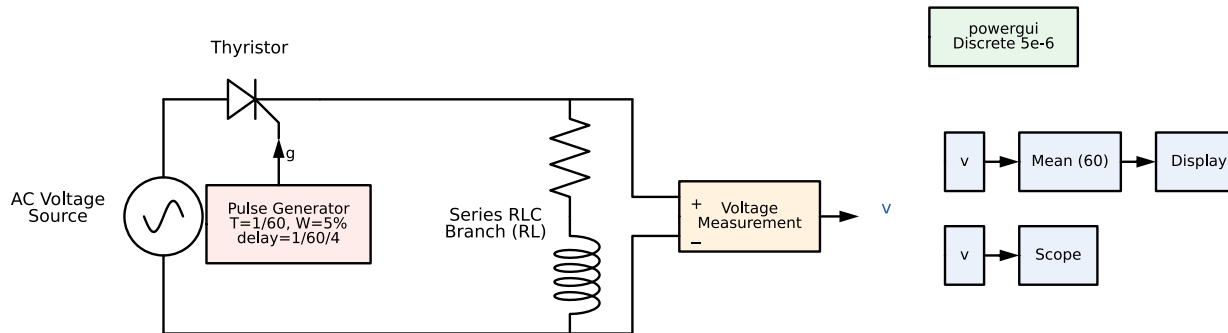
2.9 Conduction loss of a diode



Block (ADD)	SET
(circuit of 2.7)	same settings
Demux	2 on D1 m-port
Product	$i \times v$
Mean	50
Display	"Loss"
powergui	Discrete, 1e-6

Expected ≈ 15 W per diode

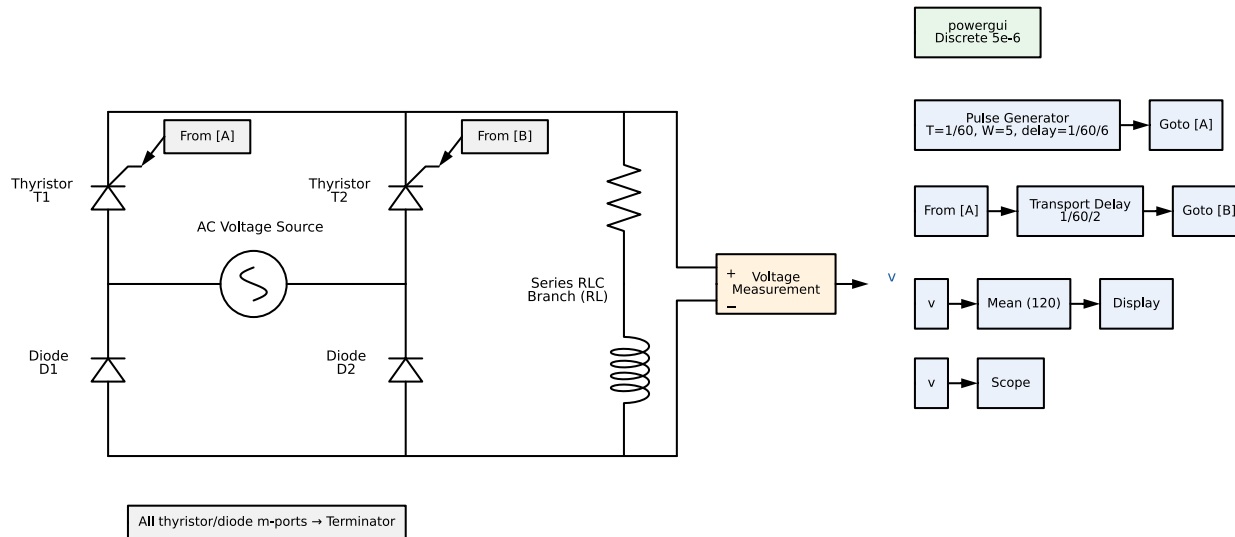
3.1 Single-phase half-wave thyristor rectifier



Block (ADD)	SET
AC Voltage Source	Peak amplitude 169.7, Phase 0, Frequency 60
Thyristor	Snubber R_s 1e7
Series RLC Branch	RL, R 10, L 10e-3
Pulse Generator	Time based, Amp 1, Period 1/60, Width 5, Phase delay 1/60/4 ($\alpha = 90^\circ$)
Voltage Measurement	—
Mean	60
Display, Scope	—
powergui	Discrete, 5e-6
Terminator	on unused m-ports

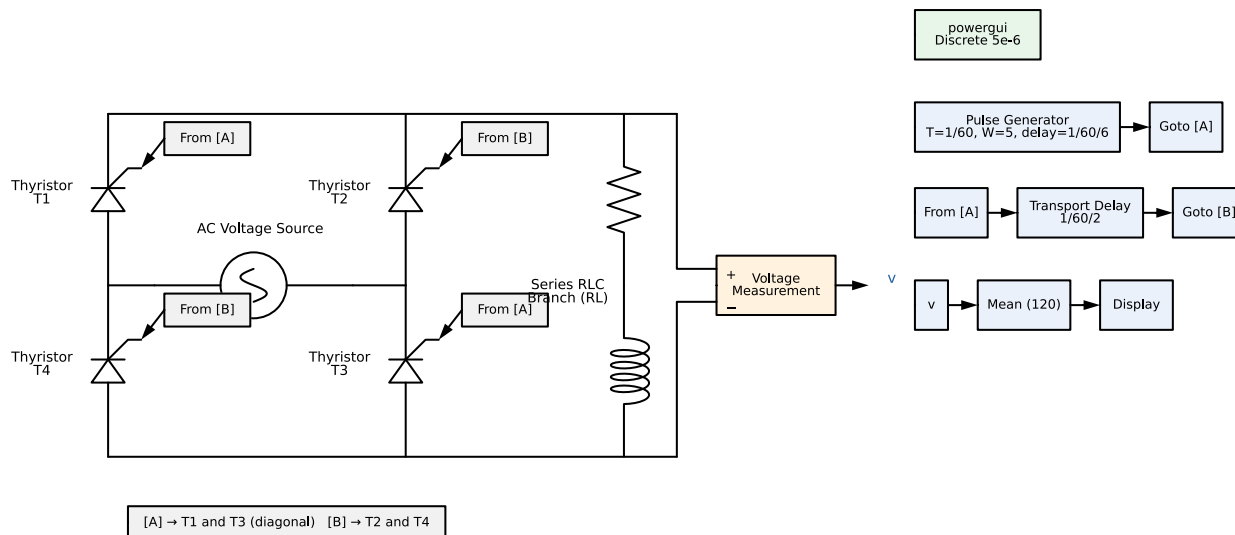
Phase delay = $\alpha/(360 \cdot f)$

3.2 Full-wave thyristor rectifier (I): 2 SCR + 2 diodes



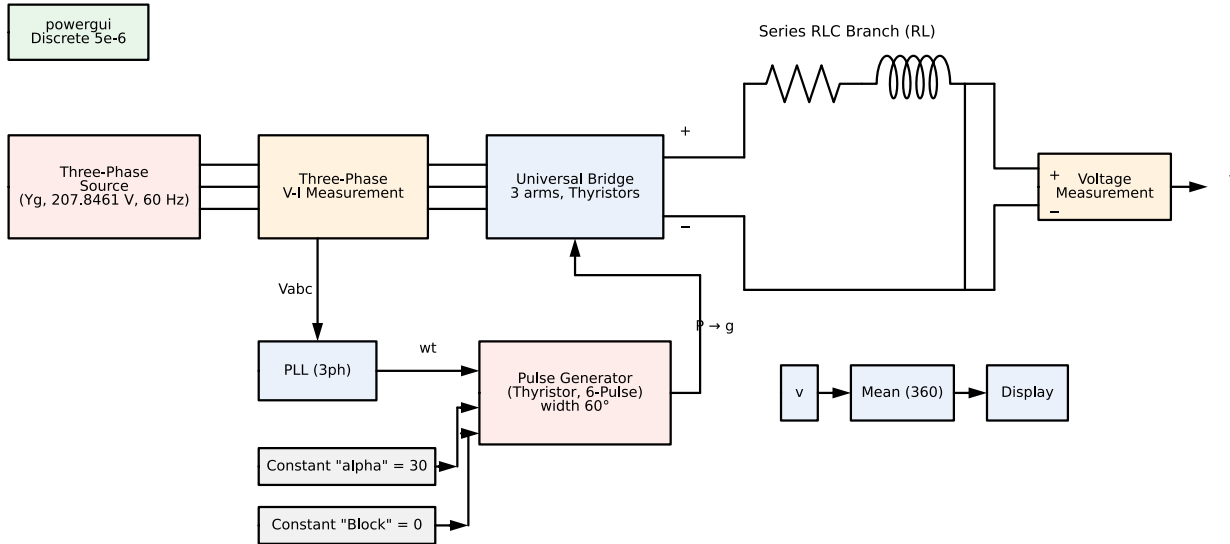
Block (ADD)	SET
AC Voltage Source	Peak amplitude 169.7 , Phase 0, Frequency 60
Thyristor ×2 (T1, T2)	Snubber Rs 1e7
Diode ×2 (D1, D2)	Snubber Rs 1e7
Series RLC Branch	RL, 10 , 10e-3
Pulse Generator	Period 1/60 , Width 5 , Phase delay 1/60/6 ($\alpha = 60^\circ$)
Transport Delay	Time delay 1/60/2
Goto / From	tags A , B (or direct wires)
Voltage Measurement	—
Mean	120
Display, Scope	—
powergui	Discrete, 5e-6
Terminator	on unused m-ports

3.3 Full-wave thyristor rectifier (II): 4 SCR



Block (ADD)	SET
AC Voltage Source	Peak amplitude 169.7 , Phase 0, Frequency 60
Thyristor ×4	Snubber Rs 1e7
Series RLC Branch	RL, 10 , 10e-3
Pulse Generator	Period 1/60 , Width 5 , delay 1/60/6
Transport Delay	1/60/2
Goto ×2 / From ×4	[A] → T1, T3; [B] → T2, T4
Voltage Measurement	—
Mean	120
powergui	Discrete, 5e-6
Terminator	on unused m-ports

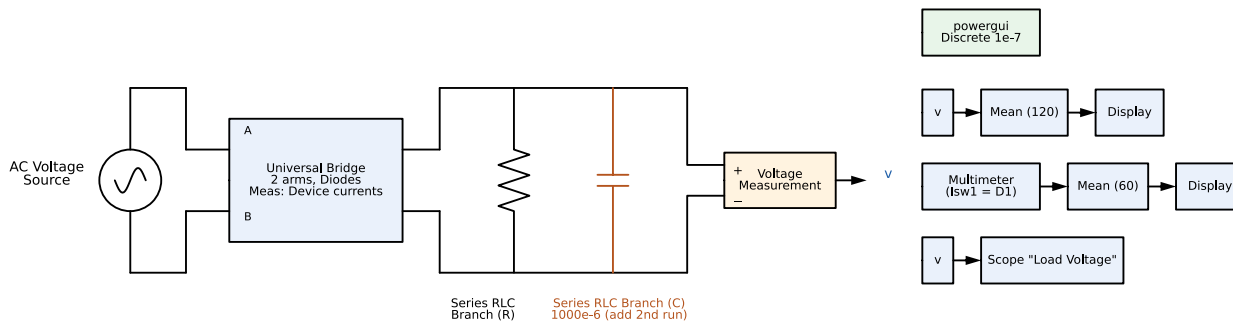
3.4 Three-phase thyristor rectifier



Block (ADD)	SET
Three-Phase Source	Yg, 207.8461 V _{LL,rms} , 60 Hz, ☐ Internal
Three-Phase V-I Measurement	phase-to-ground, current yes
Universal Bridge	arms 3, Rs 1e7, Cs inf, Thyristors, Ron 1e-3, Lon 0, Vf 0
Series RLC Branch	RL, 10, 10e-3
PLL (3ph)	defaults (45, [0 60], [180 3200 1], ☒ AGC)
Pulse Generator (Thyristor, 6-Pulse)	6-pulse, width 60°, ☐ double pulsing
Constant ×2	alpha 30; Block 0
Voltage Measurement	—
Mean	360
powergui	Discrete, 5e-6

Expected $\alpha=0^\circ$: 280.6 V, $\alpha=30^\circ$: 242.5 V

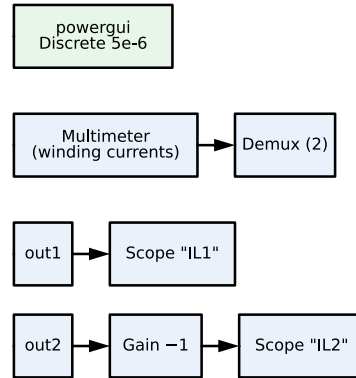
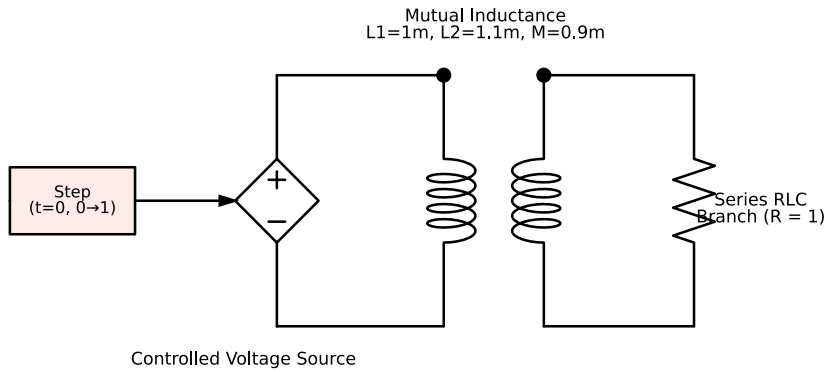
3.5 Effect of a filter capacitor



Block (ADD)	SET
AC Voltage Source	Peak amplitude 100, Phase 0, Frequency 60
Universal Bridge	arms 2, Diodes, Rs 1e7, Cs inf, Ron 1e-3, Vf 0.8, Measurements Device currents
Series RLC Branch (load)	R 10
Series RLC Branch (filter)	C 1000e-6 (2nd run), Meas: Branch current
Voltage Measurement	—
Multimeter	Isw1 (diode D1 current)
Mean ×2	120 (V _O), 60 (i _{D1})
Display, Scope	—
powergui	Discrete, 1e-7

Expected no C: 62.06 V; with C: 79.17 V, i_D peak ≈ 39 A

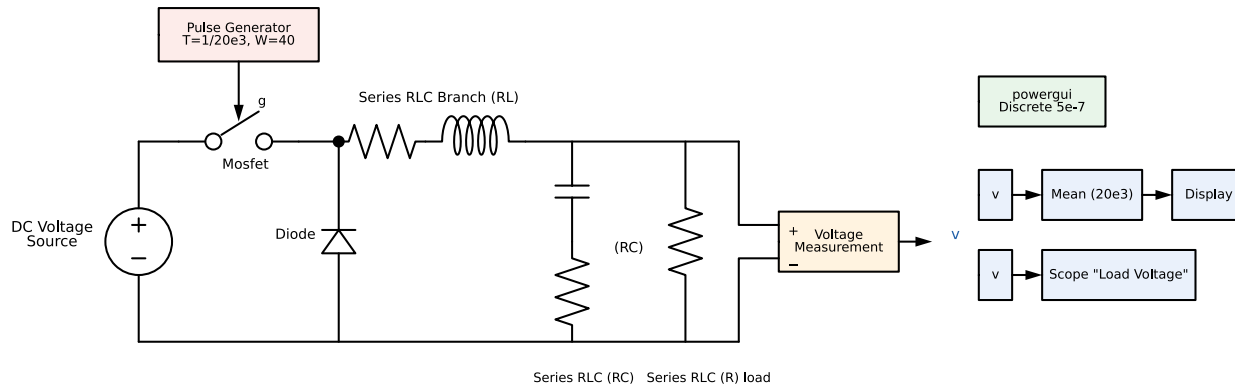
3.6 Coupled inductors



Block (ADD)	SET
Step	Step time 0 , Initial 0, Final 1
Controlled Voltage Source	defaults
Mutual Inductance	W1 [0 1e-3] , W2 [0 1.1e-3] , Mutual [0 0.9e-3] , Meas: Winding currents
Series RLC Branch	R 1
Multimeter + Demux	2 winding currents
Gain	-1 (IL2)
Scope x2	"IL1", "IL2"
powergui	Discrete, 5e-6

Stop time 2.5e-3 · IL2 → 0.9 A

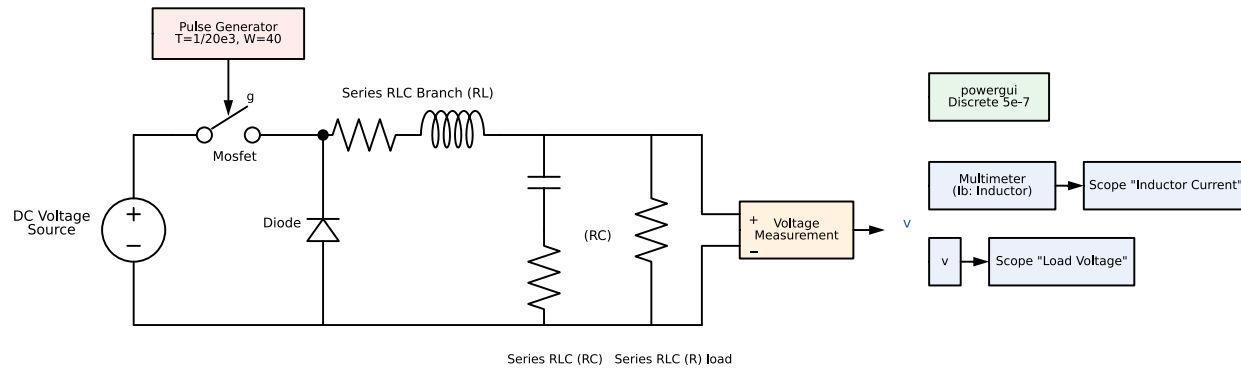
4.1 Buck converter (open loop)



Block (ADD)	SET
DC Voltage Source	Amplitude 50
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL: R 1e-3 , L 400e-6
Series RLC Branch (capacitor)	RC: R 50e-3 , C 100e-6
Series RLC Branch (load)	R 20
Pulse Generator	Amp 1 , Period 1/20e3 , Width 40 , Delay 0
Voltage Measurement	—
Mean	20e3
Display, Scope	—
powergui	Discrete, 5e-7
Terminator	on unused m-ports

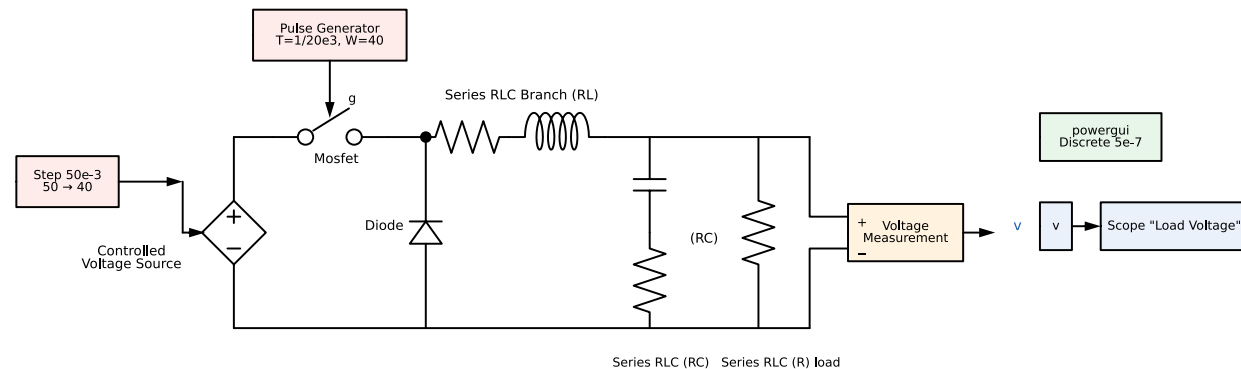
Stop time 0.05 · $V_o \approx 20$ V, ripple ≈ 100 mV

4.2 Operating mode of the converter (CCM / DCM)



Block (ADD)	SET
DC Voltage Source	Amplitude 50
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL 1e-3 , 400e-6 , Measurements Branch current
Series RLC Branch (capacitor)	RC: R 50e-3 , C 100e-6
Series RLC Branch (load)	R 20 (CCM) → 40 (DCM)
Pulse Generator	Amp 1 , Period 1/20e3 , Width 40 , Delay 0
Multimeter	Ib: Inductor
Scope	"Inductor Current"
powergui	Discrete, 5e-7

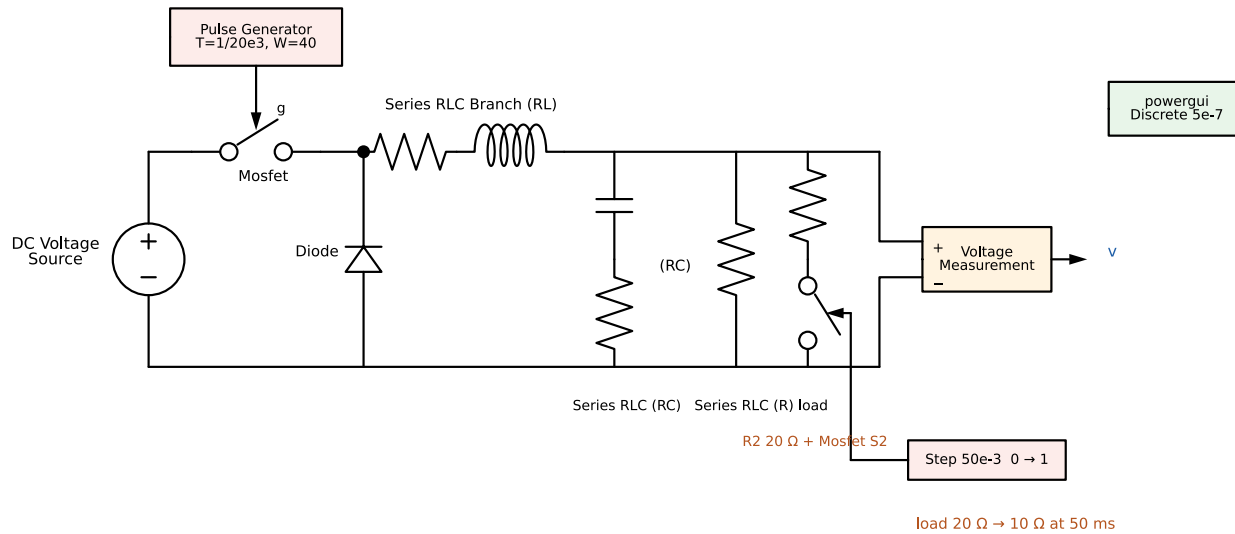
4.3 Effect of input voltage change



Block (ADD)	SET
Controlled Voltage Source	replaces DC source
Step	Step time 50e-3 , Initial 50 , Final 40 (or 60)
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL: R 1e-3 , L 400e-6
Series RLC Branch (capacitor)	RC: R 50e-3 , C 100e-6
Series RLC Branch (load)	R 20
Pulse Generator	Amp 1 , Period 1/20e3 , Width 40 , Delay 0
Voltage Measurement	—
Scope	"Load Voltage"
powergui	Discrete, 5e-7

Stop time 0.1

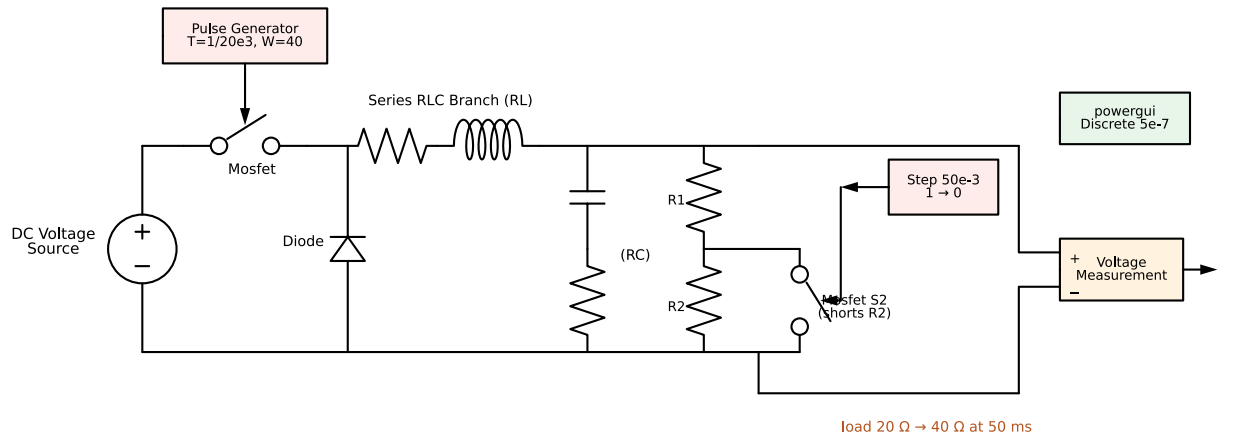
4.4a Effect of load change — decrease ($20\ \Omega \rightarrow 10\ \Omega$)



Block (ADD)	SET
DC Voltage Source	Amplitude 50
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL: R 1e-3, L 400e-6
Series RLC Branch (capacitor)	RC: R 50e-3, C 100e-6
Series RLC Branch (load)	R 20
Pulse Generator	Amp 1, Period 1/20e3, Width 40, Delay 0
Series RLC Branch "R2"	R 20
Mosfet "S2"	defaults (as a switch)
Step	time 50e-3, initial 0, final 1 $\rightarrow$ S2 g
Voltage Measurement	—
Scope	"Load Voltage"
powergui	Discrete, 5e-7

Stop time 0.1

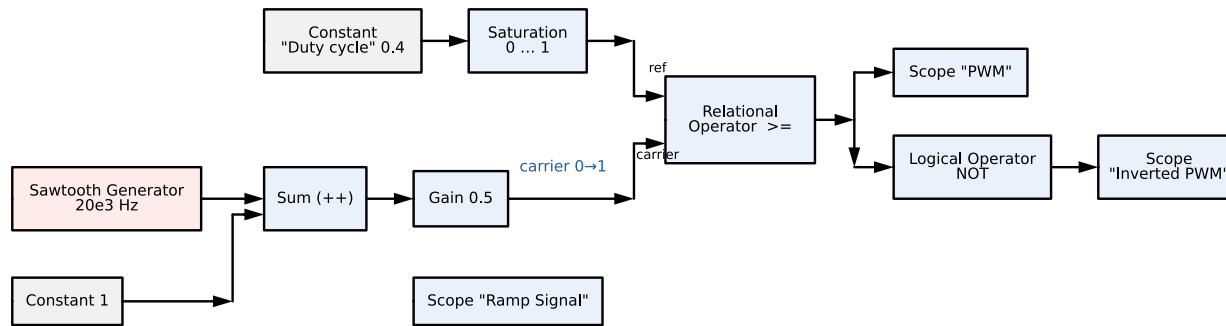
4.4b Effect of load change — increase ($20\ \Omega \rightarrow 40\ \Omega$)



Block (ADD)	SET
DC Voltage Source	Amplitude 50
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL: R 1e-3, L 400e-6
Series RLC Branch (capacitor)	RC: R 50e-3, C 100e-6
Series RLC Branch R1, R2	R 20 each, in series
Mosfet "S2"	across R2
Step	time 50e-3, initial 1, final 0 $\rightarrow$ S2 g
Pulse Generator	Amp 1, Period 1/20e3, Width 40, Delay 0
Voltage Measurement	—
powergui	Discrete, 5e-7

Stop time 0.1

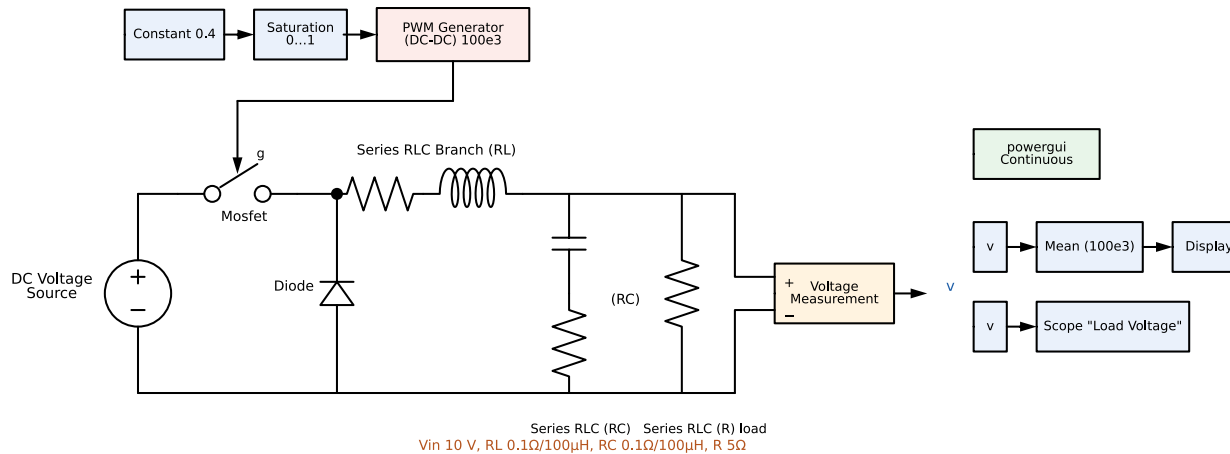
4.5 Generation of PWM signals



Block (ADD)	SET
Sawtooth Generator	Frequency 20e3 , Phase 0 (output -1...1)
Constant	1
Sum	List of signs ++
Gain	0.5 (carrier 0...1)
Constant "Duty cycle"	0.4
Saturation	Upper 1 , Lower 0
Relational Operator	>= (top = ref, bottom = carrier)
Logical Operator	NOT
Scope x3	PWM, Inverted PWM, Ramp Signal

Stop time 5e-4 · no powergui needed

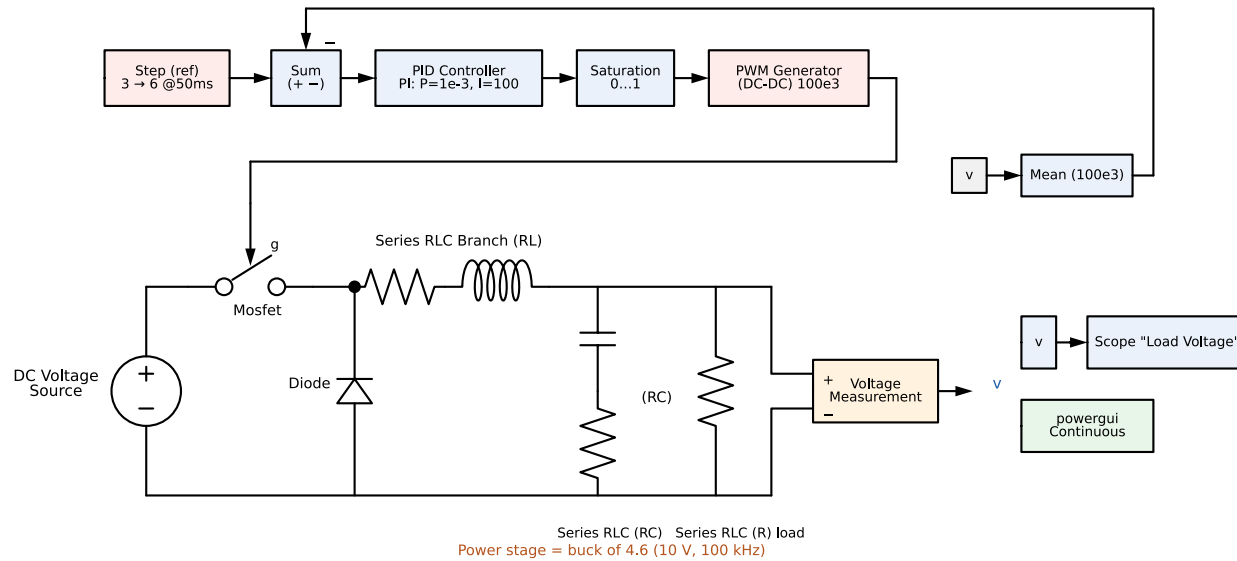
4.6 PWM Generator (DC-DC) block



Block (ADD)	SET
DC Voltage Source	10
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (inductor)	RL 0.1 , 100e-6
Series RLC Branch (capacitor)	RC 0.1 , 100e-6
Series RLC Branch (load)	R 5
Constant	0.4
Saturation	0 ... 1
PWM Generator (DC-DC)	Switching frequency 100e3
Voltage Measurement	—
Mean	100e3
powergui	Continuous

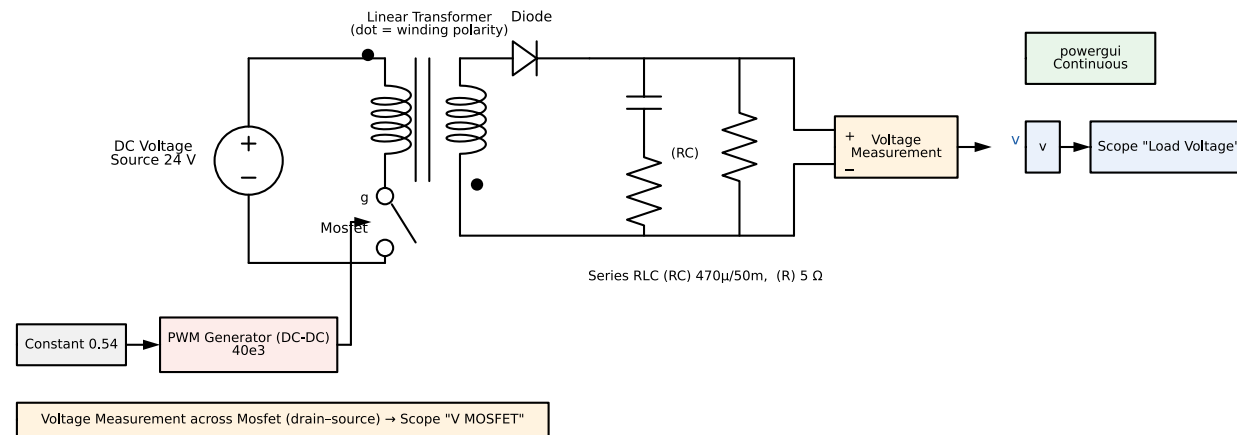
Expected 3.411–3.436 V, ripple 25 mV

4.7 Closed-loop control of a buck converter



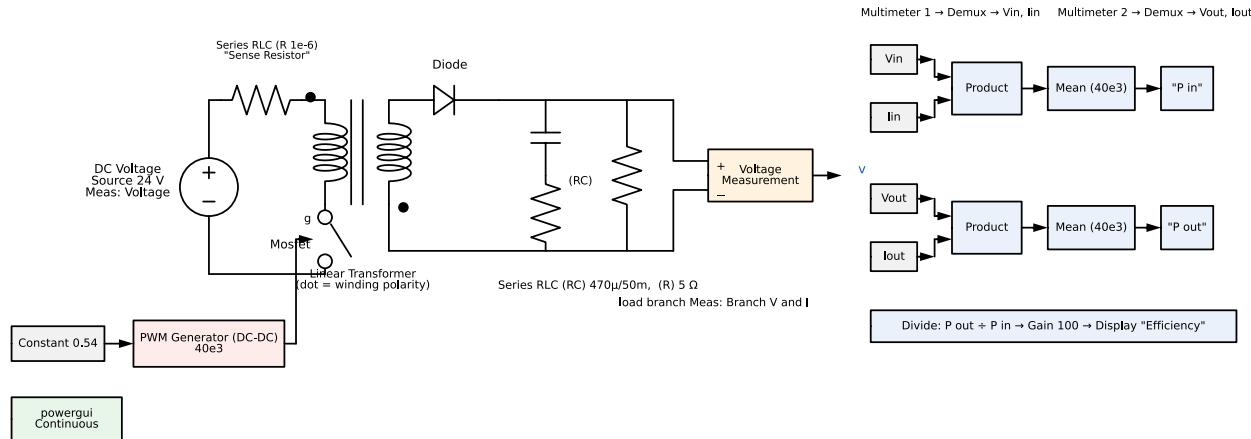
Block (ADD)	SET
(power stage of 4.6)	same settings
Step (reference)	time 50e-3 , initial 3 , final 6
Sum	List of signs + -
PID Controller	Controller PI, P 1e-3 , I 100
Saturation	0 ... 1
PWM Generator (DC-DC)	100e3
Mean (feedback)	100e3
Scope x2	Load Voltage, Duty Cycle
powergui	Continuous
Stop time 0.1	

4.8 Flyback converter



Block (ADD)	SET
DC Voltage Source	24
Linear Transformer	SI, [5 40e3] , W1 [24 0 0] , W2 [5 0 0] , □ three windings, [Rm Lm] [1e7 500e-6]
Mosfet	defaults
Diode	Snubber Rs 1e7
Series RLC Branch (capacitor)	RC 50e-3 , 470e-6
Series RLC Branch (load)	R 5
Constant	0.54
PWM Generator (DC-DC)	40e3
Voltage Measurement x2	load; Mosfet drain-source
Scope x2	Load Voltage, V MOSFET
powergui	Continuous
Stop time 0.05 · $V_{DS} \approx 52$ V steady	

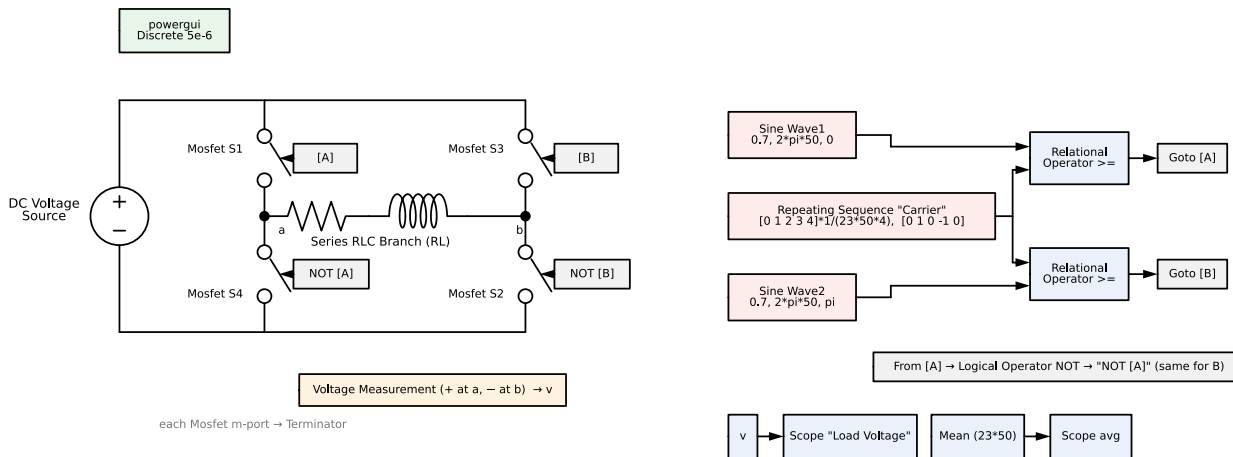
4.9 Efficiency of the flyback converter



Block (ADD)	SET
(flyback of 4.8)	DC source Measurements Voltage ; load Measurements Branch voltage and current
Series RLC Branch "Sense Resistor"	R 1e-6, Measurements Branch current
Multimeter ×2 + Demux ×2	(Vin, Iin) and (Vout, Iout)
Product ×2	—
Mean ×2	40e3
Divide	$P_{out} \div P_{in}$
Gain	100
Display	"Efficiency"
powergui	Continuous

Expected $\eta \approx 84.45\%$

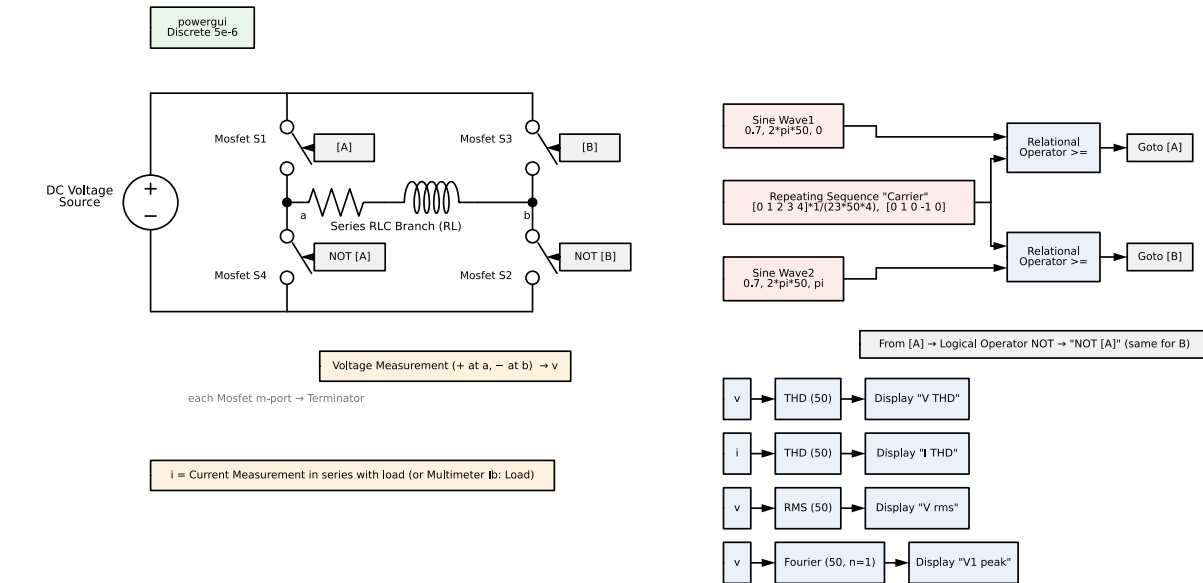
5.1 Single-phase PWM inverter (unipolar)



Block (ADD)	SET
DC Voltage Source	100
Mosfet ×4 (S1, S4, S3, S2)	defaults
Series RLC Branch (load)	RL 25, 1e-3, Meas: Branch voltage and current
Repeating Sequence "Carrier"	Time [0 1 2 3 4]*1/(23*50*4), Output [0 1 0 -1 0]
Sine Wave1 / Sine Wave2	Amp 0.7, Freq 2*pi*50, Phase 0 / pi
Relational Operator ×2	>=
Logical Operator ×2	NOT
Goto / From	A, B
Voltage Measurement	—
Mean	23*50
Scope ×2	Load Voltage, Average
powergui	Discrete, 5e-6
Terminator	on unused m-ports

Stop time 0.1 · $V_{1,peak}$ 69.32 V, V_{rms} 66.2 V

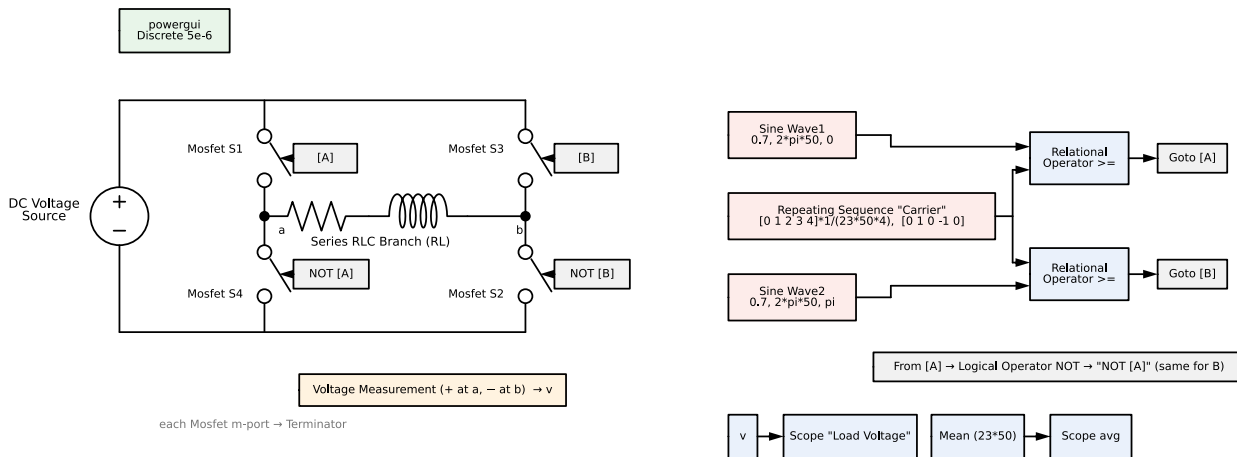
5.2 THD block



Block (ADD)	SET
(inverter of 5.1)	same settings
Current Measurement	in series with load
THD ×2	Fundamental frequency 50
RMS	50
Fourier	50, n 1
Display ×4	V THD, I THD, V rms, V1 peak
powergui	Discrete, 5e-6

Expected V-THD 90.78%, I-THD 69.4%

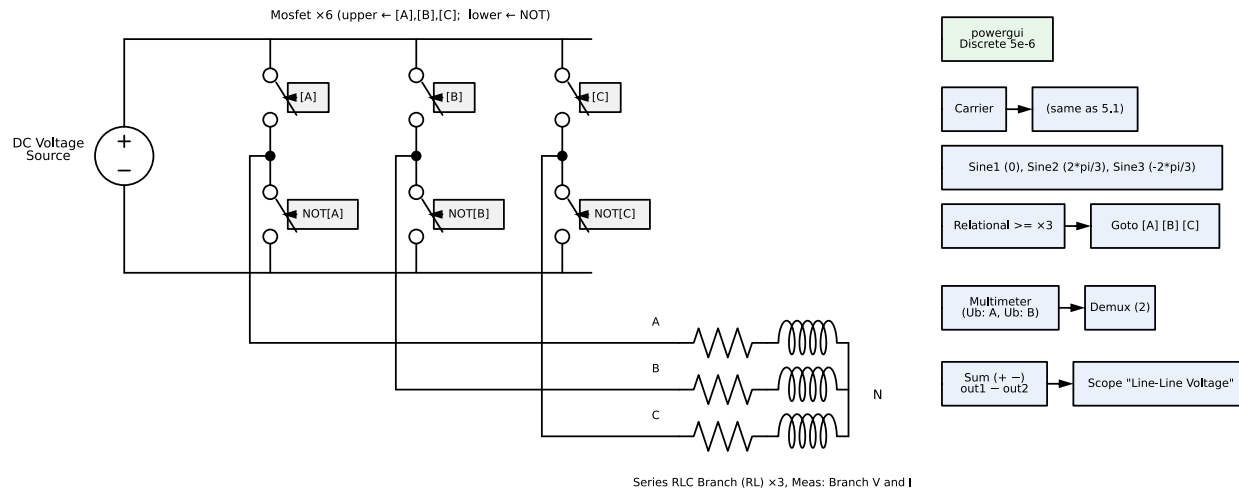
5.3 Harmonic analysis with the FFT Analyzer



Block (ADD)	SET
(inverter of 5.1)	same settings
Scope "Load Voltage" → Logging	☒ Log data to workspace, name LoadVoltage, Structure With Time
Command window	LoadVoltage = out.LoadVoltage;
powergui → Tools → FFT Analysis	Start 0.08, cycles 1, fundamental 50, Max freq 1250 (then 2500)
Display style	List (relative to specific base), Base 1

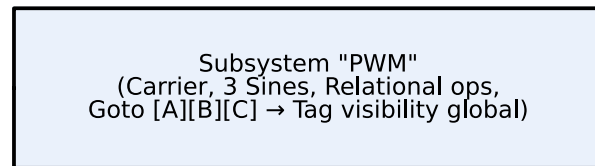
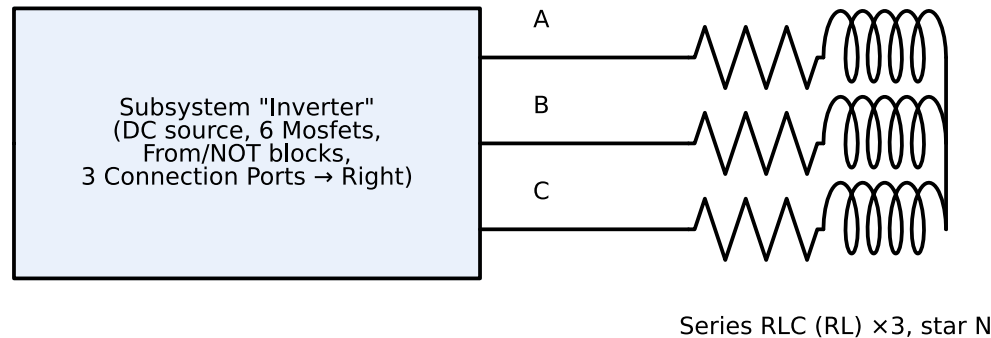
Expected dominant harmonics 43, 45, 47, 49

5.4 Three-phase PWM inverter



Block (ADD)	SET
DC Voltage Source	100
Mosfet ×6	defaults
Series RLC Branch ×3	RL 25 , 1e-3 , Meas: Branch voltage and current (star N)
Repeating Sequence	same as 5.1
Sine Wave ×3	Amp 0.7 , 2* π *50 , Phase 0 , 2* π /3 , -2* π /3
Relational Operator ×3	$\geq$
Logical Operator ×3	NOT
Goto / From	A , B , C
Multimeter + Demux + Sum	Ub:A, Ub:B → Sum +- → V_{AB}
powergui	Discrete, 5e-6
Terminator	on unused m-ports

5.5 Connection Port block (subsystem with electrical ports)



Block (ADD)	SET
(circuit of 5.4)	same settings
Connection Port ×3	Port location on parent subsystem Right
Select power stage + Ctrl+G	Subsystem "Inverter"
Goto (PWM section)	Tag visibility global
Series RLC Branch ×3	connect to subsystem terminals A, B, C

All blocks used — Block name → SET

Block name	SET
powergui	Solver → Discrete, $5e-6$ (3- ϕ rectifier $1e-6$, filter-C $1e-7$, buck $5e-7$); PWM Generator / flyback → Continuous
AC Voltage Source	Peak amplitude (= $V_{rms} \times \sqrt{2}$), Phase (deg), Frequency (Hz)
DC Voltage Source	Amplitude (V); Measurements <i>Voltage</i> if needed
Controlled Voltage Source	input from Step / signal
Three-Phase Source	Yg, Phase-to-phase Vrms, Freq, ☐ Internal
Diode	Snubber Rs $1e7$ (others default: Ron $1e-3$, Vf 0.8)
Thyristor	Snubber Rs $1e7$; gate g ← Pulse Generator
Mosfet	defaults; gate g ← Pulse / PWM
Universal Bridge	arms 2 or 3, Rs $1e7$, Cs inf, device Diodes / Thyristors, Ron $1e-3$
Series RLC Branch	Branch type R / L / C / RL / RC; values; Measurements (Branch voltage / current)
Three-Phase Series RLC Branch	type RL, R, L
Mutual Inductance	[R1 L1], [R2 L2], [Rm Lm]
Linear Transformer	[Pn fn], W1 [V R L], W2 [V R L], [Rm Lm]
Voltage Measurement	+ / - across element → v
Current Measurement	in series → i
Multimeter	choose Ub / Ib / Isw from list → >>
Three-Phase V-I Measurement	phase-to-ground, current yes
Mean	Fundamental frequency: f (HW), 2f (FW), 6f (3- ϕ), f_s (DC-DC)
RMS	Fundamental frequency (as Mean)
Fourier	Fundamental f, Harmonic n (e.g. 0:5), Initial [0,0]
THD	Fundamental frequency (output f)
Power	Fundamental frequency (linear circuits only)
Power (3ph, Instantaneous)	—
PLL (3ph)	defaults
Pulse Generator (Thyristor, 6-Pulse)	6-pulse, width 60°

Block name	SET
Pulse Generator	Time based, Amp 1, Period $1/f$, Width (%), Phase delay $\alpha / (360 \cdot f)$
PWM Generator (DC-DC)	Switching frequency; input D (0...1)
Sawtooth Generator	Frequency; output -1...1
Repeating Sequence	Time [0 1 2 3 4] * $1 / (m_f \cdot f \cdot 4)$, Output [0 1 0 -1 0]
Sine Wave	Amplitude m_a , Frequency $2 \cdot \pi \cdot f$ (rad/s), Phase (rad)
Transport Delay	Time delay $1 / (2f)$
Step	Step time, Initial value, Final value
Constant	value (α , D, 0, 1...)
Saturation	Upper 1, Lower 0
Relational Operator	>=
Logical Operator	NOT
Sum	List of signs ++ / +- -
Gain	value (0.5, 3, -1, 100)
Product	Number of inputs 2
Divide	× top, ÷ bottom
PID Controller	PI, P, I
Demux	Number of outputs
Mux	Number of inputs
Goto / From	tag; Tag visibility global for subsystems
Connection Port	Port location Right
Scope / Display	— (rename to what it shows)
Terminator	on unused m-ports / outputs
Ground	star point of 3- ϕ sources